Assignment

SQL Case Study – 2 Simple Queries:

1. List all the employee details

```
SELECT * FROM EMPLOYEE;
```

2. List all the department details.

```
SELECT * FROM DEPARTMENT;
```

3. List all job details.

```
SELECT * FROM JOB;
```

4. List all the locations.

```
SELECT * FROM LOCATION;
```

5. List out the First Name, Last Name, Salary, Commission for all Employees.

```
SELECT FIRST_NAME,
LAST_NAME,
SALARY,
COMM
 FROM EMPLOYEE;
```

6. Display Employee ID, Last Name and Department ID using aliases.

```
SELECT EMPLOYEE_ID AS "ID of the Employee",
LAST_NAME AS "Name of the Employee",
DEPARTMENT_ID AS "Dep_id"
 FROM EMPLOYEE;
```

7. List out the annual salary of the employees with their names only.

```
SELECT FIRST_NAME,
LAST_NAME,
SALARY * 12 AS Annual_Salary
 FROM EMPLOYEE;
```

WHERE Condition:

1. List the details about "Smith".

```
SELECT *
FROM EMPLOYEE
WHERE LAST_NAME = 'SMITH';
```

2. List out the employees who are working in department 20.

```
SELECT *
FROM EMPLOYEE
WHERE DEPARTMENT_ID = 20;
```

3. List out the employees who are earning salary between 2000 and 3000

```
SELECT *
FROM EMPLOYEE
WHERE SALARY BETWEEN 2000 AND 3000;
```

4. List out the employees who are working in department 10 or 20.
```
SELECT *
FROM EMPLOYEE
WHERE DEPARTMENT_ID IN (10,20);
```
5. Find out the employees who are not working in department 10 or 30.
```
SELECT *
FROM EMPLOYEE
WHERE DEPARTMENT_ID NOT IN (10,30);
```
6. List out the employees whose name starts with 'L'.
```
SELECT *
FROM EMPLOYEE
WHERE FIRST_NAME LIKE 'L%';
```
7. List out the employees whose name starts with 'L' and ends with 'E'.
```
SELECT *
FROM EMPLOYEE
WHERE FIRST_NAME LIKE 'L%E';
```
8. List out the employees whose name length is 4 and start with 'J'.
```
SELECT *
FROM EMPLOYEE
WHERE FIRST_NAME LIKE 'J___';
```
9. List out the employees who are working in department 30 and draw the salaries more than 2500.
```
SELECT *
FROM EMPLOYEE
WHERE DEPARTMENT_ID = 30
AND SALARY > 2500;
```
10. List out the employees who are not receiving commission.
```
SELECT *
FROM EMPLOYEE
WHERE COMM IS NULL;
```

ORDER BY Clause:

1. List out the Employee ID and Last Name in ascending order based on the Employee ID.
```
SELECT EMPLOYEE_ID,
LAST_NAME
FROM EMPLOYEE
ORDER BY EMPLOYEE_ID ASC;
```
2. List out the Employee ID and Name in descending order based on salary.

```
SELECT EMPLOYEE_ID,
FIRST_NAME
FROM EMPLOYEE
ORDER BY SALARY DESC;
```

3.List out the employee details according to their Last Name in ascending-order.

```
SELECT *
FROM EMPLOYEE
ORDER BY LAST_NAME ASC;
```

4. List out the employee details according to their Last Name in ascending order and then Department ID in descending order.

```
SELECT *
FROM EMPLOYEE
ORDER BY LAST_NAME ASC,
DEPARTMENT_ID DESC;
```

GROUP BY and HAVING Clause:

1. List out the department wise maximum salary, minimum salary and average salary of the employees.
```
SELECT DEPARTMENT_ID,
MAX(SALARY) AS MAX_SALARY,
MIN(SALARY) AS MIN_SALARY,
AVG(SALARY) AS AVG_SALARY
FROM EMPLOYEE
GROUP BY DEPARTMENT_ID;
```
2. List out the job wise maximum salary, minimum salary and average salary of the Employees
```
SELECT JOB_ID,
MAX(SALARY) AS MAX_SALARY,
MIN(SALARY) AS MIN_SALARY,
AVG(SALARY) AS AVG_SALARY
FROM EMPLOYEE
GROUP BY JOB_ID;
```
3. List out the number of employees who joined each month in ascending order.
```
SELECT MONTHNAME(HIRE_DATE) AS MONTH,
COUNT(*) AS NO_OF_EMPLOYEES
FROM EMPLOYEE
GROUP BY MONTH(HIRE_DATE), MONTHNAME(HIRE_DATE)
ORDER BY MONTH(HIRE_DATE);
```
4. List out the number of employees for each month and year in ascending order based on the year and month.

```
SELECT YEAR(HIRE_DATE) AS YEAR,
MONTHNAME(HIRE_DATE) AS MONTH,
COUNT(*) AS NO_OF_EMPLOYEES
FROM EMPLOYEE
GROUP BY YEAR(HIRE_DATE), MONTH(HIRE_DATE), MONTHNAME(HIRE_DATE)
ORDER BY YEAR(HIRE_DATE), MONTH(HIRE_DATE);
```

5. List out the Department ID having at least four employees.

```
SELECT DEPARTMENT_ID,
COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
GROUP BY DEPARTMENT_ID
HAVING COUNT(*) >= 4;
```

6. How many employees joined in February month.

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
WHERE MONTH(HIRE_DATE) = 2;
```

7. How many employees joined in May or June month.

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
WHERE MONTH(HIRE_DATE) IN (5,6);
```

8. How many employees joined in 1985?

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
WHERE YEAR(HIRE_DATE) = 1985;
```

9. How many employees joined each month in 1985?

```
SELECT MONTHNAME(HIRE_DATE) AS MONTH,
COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
WHERE YEAR(HIRE_DATE) = 1985
GROUP BY MONTH(HIRE_DATE), MONTHNAME(HIRE_DATE)
ORDER BY MONTH(HIRE_DATE);
```

10. How many employees were joined in April 1985?

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
WHERE MONTH(HIRE_DATE) = 4
AND YEAR(HIRE_DATE) = 1985;
```

11. Which is the Department ID having greater than or equal to 3 employees joining in April 1985?

```
SELECT DEPARTMENT_ID,
COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
WHERE MONTH(HIRE_DATE) = 4
```

```
AND YEAR(HIRE_DATE) = 1985
GROUP BY DEPARTMENT_ID
HAVING COUNT(*) >= 3;
```

Joins:

1. List out employees with their department names.

```
SELECT E.EMPLOYEE_ID,
E.FIRST_NAME,
E.LAST_NAME,
D.NAME AS DEPARTMENT_NAME
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

2. Display employees with their designations.

```
SELECT E.EMPLOYEE_ID,
E.FIRST_NAME,
E.LAST_NAME,
J.DESIGNATION
FROM EMPLOYEE E
JOIN JOB J
ON E.JOB_ID = J.JOB_ID;
```

3. Display the employees with their department names and city.

```
SELECT E.EMPLOYEE_ID,
E.FIRST_NAME,
E.LAST_NAME,
D.NAME AS DEPARTMENT_NAME,
L.CITY
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
JOIN LOCATION L
ON D.LOCATION_ID = L.LOCATION_ID;
```

4. How many employees are working in different departments? Display with department names.

```
SELECT D.NAME AS DEPARTMENT_NAME,
COUNT(E.EMPLOYEE_ID) AS TOTAL_EMPLOYEES
FROM EMPLOYEE E
JOIN DEPARTMENT D
```

```sql
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
GROUP BY D.NAME;
```

5. How many employees are working in the sales department?

```sql
SELECT COUNT(E.EMPLOYEE_ID) AS TOTAL_EMPLOYEES
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
WHERE D.NAME = 'Sales';
```

6. Which is the department having greater than or equal to 3 employees and display the department names in ascending order.

```sql
SELECT D.NAME AS DEPARTMENT_NAME,
    COUNT(E.EMPLOYEE_ID) AS TOTAL_EMPLOYEES
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
GROUP BY D.NAME
HAVING COUNT(E.EMPLOYEE_ID) >= 3
ORDER BY D.NAME ASC;
```

7. How many employees are working in 'Dallas'?

```sql
SELECT COUNT(E.EMPLOYEE_ID) AS TOTAL_EMPLOYEES
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
JOIN LOCATION L
ON D.LOCATION_ID = L.LOCATION_ID
WHERE L.CITY = 'Dallas';
```

8. Display all employees in sales or operation departments.

```sql
SELECT E.EMPLOYEE_ID,
E.FIRST_NAME,
E.LAST_NAME,
D.NAME AS DEPARTMENT_NAME
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
WHERE D.NAME IN ('Sales', 'Operations');
```

CONDITIONAL STATEMENT

1. Display the employee details with salary grades. Use conditional statement to create a grade column.

```
    SELECT EMPLOYEE_ID,
   FIRST_NAME,
   LAST_NAME,
   SALARY,
   CASE
     WHEN SALARY >= 4000 THEN 'A'
     WHEN SALARY >= 3000 THEN 'B'
     WHEN SALARY >= 2000 THEN 'C'
     WHEN SALARY >= 1000 THEN 'D'
     ELSE 'E'
   END AS GRADE
FROM EMPLOYEE;
```

2. List out the number of employees grade wise. Use conditional statement to create a grade column.

```
SELECT
  CASE
    WHEN SALARY >= 4000 THEN 'A'
    WHEN SALARY >= 3000 THEN 'B'
    WHEN SALARY >= 2000 THEN 'C'
    WHEN SALARY >= 1000 THEN 'D'
    ELSE 'E'
  END AS GRADE,
  COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
GROUP BY
  CASE
    WHEN SALARY >= 4000 THEN 'A'
    WHEN SALARY >= 3000 THEN 'B'
    WHEN SALARY >= 2000 THEN 'C'
    WHEN SALARY >= 1000 THEN 'D'
    ELSE 'E'
  END;
```

3. Display the employee salary grades and the number of employees between 2000 to 5000 range of salary.

```
   SELECT
  CASE
```

```
        WHEN SALARY >= 4000 THEN 'A'
        WHEN SALARY >= 3000 THEN 'B'
        WHEN SALARY >= 2000 THEN 'C'
    END AS GRADE,
    COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEE
WHERE SALARY BETWEEN 2000 AND 5000
GROUP BY
    CASE
        WHEN SALARY >= 4000 THEN 'A'
        WHEN SALARY >= 3000 THEN 'B'
        WHEN SALARY >= 2000 THEN 'C'
    END;
```